

LECTURE 23 SCRIBE

Predicting the Impossible: Tail Bounds and System Reliability

Case Study: Amazon Black Friday Sale 2026

Slide 1: Lecture Title

Instructor:

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Slide 2: Amazon Black Friday Sale 2026

Details:

- Expected Start Date: 27 November 2026 (Friday)
- Based on historical patterns and forecasts

Observation:

- Large-scale uncertainty
- Heavy traffic and unpredictable user behavior

Slide 3: Uncertainty in Computing Systems

- Real-world systems operate under uncertainty
- User requests and traffic are random

Challenge: Exact prediction is not possible.

Slide 4: Motivation

- Handle unpredictable system behavior
- Estimate extreme conditions like overload

Approach: Use probabilistic bounds.

Slide 5: Tail Bounds

Definition: Tail bounds give an upper bound on probability that a random variable deviates significantly from its expected value.

Purpose:

- Quantify risk of extreme events
- Assist in system reliability

Slide 6: Types of Tail Bounds

1. Markov's Inequality
2. Chebyshev's Inequality
3. Chernoff Bound

Slide 7: Markov's Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Assumptions:

- $X \geq 0$
- $E[X]$ exists

Slide 8: Derivation of Markov's Inequality

$$E[X] = \int_0^\infty P(X \geq t) dt$$

For $t \leq a$:

$$P(X \geq t) \geq P(X \geq a)$$

$$E[X] \geq \int_0^a P(X \geq a) dt = a \cdot P(X \geq a)$$

$$\Rightarrow P(X \geq a) \leq \frac{E[X]}{a}$$

Slide 9: Interpretation of Markov's Inequality

- Requires only mean
- Provides loose bound
- Useful with limited information

Slide 10: Chebyshev's Inequality

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Assumptions:

- Mean μ exists
- Variance σ^2 exists

Slide 11: Derivation of Chebyshev's Inequality

Apply Markov to $(X - \mu)^2$:

$$P((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{E[(X - \mu)^2]}{k^2\sigma^2}$$

Since:

$$E[(X - \mu)^2] = \sigma^2$$

$$\Rightarrow P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Slide 12: Interpretation of Chebyshev's Inequality

- Works for any distribution
- Uses variance
- Tighter than Markov

Slide 13: Chernoff Bound

$$P(X \geq (1 + \delta)\mu) \leq e^{-\frac{\mu\delta^2}{3}}, \quad (0 < \delta < 1)$$

Assumptions:

- Independent random variables
- Often Bernoulli trials

Slide 14: Properties of Chernoff Bound

- Exponentially decreasing bounds
- Much tighter
- Suitable for large-scale systems

Slide 15: Application to Amazon Case Study

Model:

- X = system load (number of users)

Objective:

- Estimate probability of overload and failure

Method: Apply tail bounds.

Slide 16: System Reliability

Definition: Probability that a system performs correctly under specified conditions.

Use:

- Estimate failure probability
- Design safety margins

Slide 17: Key Observations

Method	Requirement	Nature
Markov	Mean only	Loose
Chebyshev	Mean + Variance	Moderate
Chernoff	Independence	Tight

- Exact prediction not feasible
- Tail bounds provide upper guarantees

Slide 18: Class Participation

- Discussion on tail bounds application
- System reliability
- Interpretation of probabilistic guarantees